Charis Xiong

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Education

Rice University **Houston, TX**
Bachelor of Science in Statistics | Minors: Data Science, Asian Studies
- **Expected graduation**: December 2026| **GPA:** 3.91
- **Coursework:** DS Tools and Models; Probability; R for DS; Practical Machine Learning; Data Visualization
- **Leadership Roles:** Peer Academic Advisor (PAA); General Education Student Ambassador, Teaching Assistant
- **Awards**: 1st in DEEP (2025), Regeneron ISEF 2023 – 3rd Grand Award (Plant Sciences), National Geographic Award (2023)

Professional and Research Experience

NSA 2026 Future Computing Summer Internship **Baltimore, MD**
Summer Research Intern *May 2026 – Current*
- Selected for the Cognitive Paradigm: AI Project to Explore Symbolic Computing, investigating symbolic computing as a brain-inspired approach to artificial intelligence and future computing.
- Apply Python, modeling, and simulation skills to explore how symbolic systems can represent and process information, drawing on experience in statistics, AI/ML, Linux, and computer architecture.

UT MD Anderson Cancer Center – Department of Epidemiology **Houston, TX**
Undergraduate Researcher (Causal Inference & Genomic Software Development) *November 2025 – Current*
- Contribute to a simulation framework for maternal–fetal health genomics using generative models to capture mother–child–placenta data patterns and evaluate causal inference assumptions.
- Create simulated datasets under different exposure and confounding scenarios to compare causal inference methods.
- Support development of an R package that recommends appropriate causal inference approaches based on simulation results.

Break Through Tech AI Program **Online**
AI/ML Fellow *May 2026 – Current*
- Selected for a competitive AI program focused on applied machine learning, model evaluation, responsible AI, and industry-connected technical projects.
- Build practical skills in data science, machine learning workflows, and technical communication through portfolio-based AI/ML work.

Rice University **Houston, TX**
Labbie/Grader *August 2024 – Current*
- Instructed students in applied statistics and data visualization using R and RStudio, covering topics like t-tests, probability distributions, and regression analysis.
- Assessed and graded assignments for students, providing clear, actionable feedback on statistical analysis and R-based coding

WorldStrides **Houston, TX**
Rice Elite Tech Camp - Live Instructor *May 2024 – July 2025*
- Taught students the fundamentals of Python programming, guiding them through key concepts such as variables, loops, functions, and data structures, while showing how coding can be applied to solve practical, real-world problems.
- Introduced core concepts in machine learning and deep learning, including model building, evaluation metrics, and applications in areas like image recognition, emphasizing both technical foundations and real-world relevance.

Projects

Data Exploration and Experimental Projects (DEEP) Mentor *August 2024-December 2025*
- Mentor student teams through complete data science workflows, offering guidance on dataset selection, data cleaning, exploratory analysis, feature engineering, model building, and evaluation using real-world datasets.
- Supported projects such as a housing market prediction analysis that won 1st place, helping students debug code, interpret models, compare metrics, and present results clearly.

BioKind Analytic Analyst – Melanin Minds Initiative *August 2025-December 2025*
- Analyze survey and event data using Python and NLP to identify key themes, sentiment patterns, and engagement trends.
- Create geographic and participation visualizations to support outreach evaluation and measure community impact.

Life Expectancy Prediction Project *January 2025-April 2025*
- Built and compared linear and polynomial regression models in R using health, demographic, and economic indicators from a Kaggle dataset, including data cleaning, feature selection, and exploratory analysis.
- Applied Box-Cox transformations, full model diagnostics (residual checks, multicollinearity tests, heteroscedasticity assessment), and model comparison metrics to evaluate performance and identify key drivers of life expectancy.

Skills

Languages: Python, R, SQL
Libraries: pandas, NumPy, matplotlib, seaborn, scikit-learn, tidyverse, ggplot2, dplyr, tidyr
Tools: Git/GitHub, Jupyter Notebook, RStudio, VS Code, Excel, Power BI, Tableau
Statistical Methods: Regression, probability distributions, hypothesis testing, bootstrapping, time series analysis
Machine Learning: Random Forest, SVM, feature engineering, cross-validation, model diagnostics
Data Engineering: AWS S3, SQLite, Hadoop basics, ETL pipelines, data cleaning, web scraping
AI/NLP: TensorFlow, Keras, text preprocessing, sentiment analysis, keyword extraction